

RISING WATERS

“Flood Prediction System Using Machine Learning”

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ABSTRACT

- Floods are among the most devastating natural disasters, causing severe damage to human lives, property, agriculture, and infrastructure.
- Accurate flood prediction is essential for minimizing risks and enabling timely disaster preparedness.
- The **Rising Waters** project is a machine learning-based flood prediction system that analyzes environmental and geographical factors such as monsoon intensity, topography drainage, river management, deforestation, urbanization, climate change, dam quality, and other flood-related parameters.
- The system uses the **K-Nearest Neighbors (KNN) Regressor** algorithm to predict the probability of flooding and classify the risk level as **Low, Moderate, or High**.
- A user-friendly web application developed using **Flask** allows users to enter environmental parameters and receive instant flood risk predictions.
- This project helps improve decision-making, supports disaster management authorities, and contributes to reducing the impact of flood-related disasters.

PROBLEM STATEMENT

- Floods are one of the most frequent and destructive natural disasters, causing significant loss of life, property, agriculture, and infrastructure.
- Existing flood prediction methods often rely on manual observations or traditional forecasting techniques, which may not provide accurate or timely warnings.
- The lack of an efficient prediction system makes it difficult for authorities and communities to take preventive measures before a flood occurs.
- The **Rising Waters** project addresses this challenge by using **Machine Learning** to analyze multiple environmental factors, including monsoon intensity, topography drainage, river management, deforestation, urbanization, climate change, dam quality, and other flood-related parameters.
- By predicting flood probability and identifying risk levels, the system enables early warnings, supports disaster management, and helps reduce the impact of flood-related disasters.

Key Points

- Delayed flood warnings increase damage and loss of life.
- Traditional prediction methods may not provide sufficient accuracy.
- Manual analysis is time-consuming and less efficient.
- Early flood prediction is essential for effective disaster preparedness.
- Machine Learning provides faster and more accurate flood risk assessment.

FEATURES

-  **Flood Probability Prediction** – Predicts the likelihood of flooding using the K-Nearest Neighbors (KNN) machine learning algorithm.
-  **Multiple Input Parameters** – Analyzes 20 environmental and geographical factors such as monsoon intensity, river management, urbanization, climate change, and drainage systems.
-  **Flood Risk Classification** – Classifies the predicted flood risk into **Low**, **Moderate**, and **High** levels.
-  **User-Friendly Web Interface** – Provides an easy-to-use Flask-based application for entering input values and viewing prediction results.
-  **Real-Time Prediction** – Generates instant flood probability predictions based on user inputs.
-  **Data Visualization** – Supports graphical analysis through correlation heatmaps, distribution plots, histograms, and scatter plots for better understanding of the dataset.
-  **Accurate Machine Learning Model** – Uses a trained KNN Regressor model to improve prediction accuracy.
-  **Decision Support** – Assists disaster management authorities and communities in taking timely preventive measures and improving flood preparedness.

TECH STACK

- The **Rising Waters** project is developed using modern programming tools and machine learning libraries to ensure accurate flood prediction and a user-friendly web interface.
- Python is used as the primary programming language, while Flask provides the web application framework. The K-Nearest Neighbors (KNN) algorithm is implemented using Scikit-learn to predict flood probability based on environmental factors.
- Pandas and NumPy are used for data preprocessing, and Matplotlib and Seaborn are used for data visualization.

Technologies Used

-  **Programming Language:** Python
-  **Web Framework:** Flask
-  **Machine Learning:** Scikit-learn (KNN Regressor)
-  **Data Processing:** Pandas, NumPy
-  **Model Storage:** Joblib
-  **Frontend:** HTML5, CSS3
-  **Data Visualization:** Matplotlib, Seaborn
-  **Dataset:** Flood Prediction Dataset (flood.csv)
-  **Development Tools:** Visual Studio Code (VS Code), Jupyter Notebook

INSTALLATION STEPS

- To run the **Rising Waters** project successfully, the required software and Python libraries must be installed. The application uses the Flask framework and a trained Machine Learning model to predict flood probability. Follow the steps below to set up and execute the project.

Installation Process

Step 1: Download or clone the project folder.

Step 2: Open the project in **Visual Studio Code (VS Code)**.

Step 3: Install all the required Python libraries using the following command:

```
pip install -r requirements.txt
```

Step 4: Train the Machine Learning model (if the model file is not already available):

```
python train_model.py
```

Step 5: Run the Flask application:

```
python app.py
```

Step 6: Open your web browser and visit:

```
http://127.0.0.1:5000
```

Step 7: Enter the required environmental parameters (values between 0–10) and click "**Predict Flood Probability**" to view the flood risk prediction.

Outcome

After completing these steps, the Rising Waters application will be ready to predict flood probability and display the corresponding risk level through a simple web interface.

USAGE INSTRUCTIONS

- The **Rising Waters** application provides a simple and interactive interface for predicting flood probability.
- Users only need to enter the required environmental parameters, and the system instantly analyzes the data using the trained **K-Nearest Neighbors (KNN)** model to generate the flood prediction and risk level.

Steps to Use the Application

Step 1: Launch the application by running app.py.

Step 2: Open the application in your web browser.

Step 3: Enter values (0–10) for all **20 flood-related parameters**, such as:

- Monsoon Intensity
- Topography Drainage
- River Management
- Deforestation
- Urbanization
- Climate Change

USAGE INSTRUCTIONS

- Wetland Loss
- Dams Quality
- Drainage Systems
- Political Factors

Step 4: Click the "**Predict Flood Probability**" button.

Step 5: The system processes the input using the trained **KNN Regressor** model.

Step 6: The application displays:

-  **Flood Probability**
-  **Low Risk**
-  **Moderate Risk**
-  **High Risk**

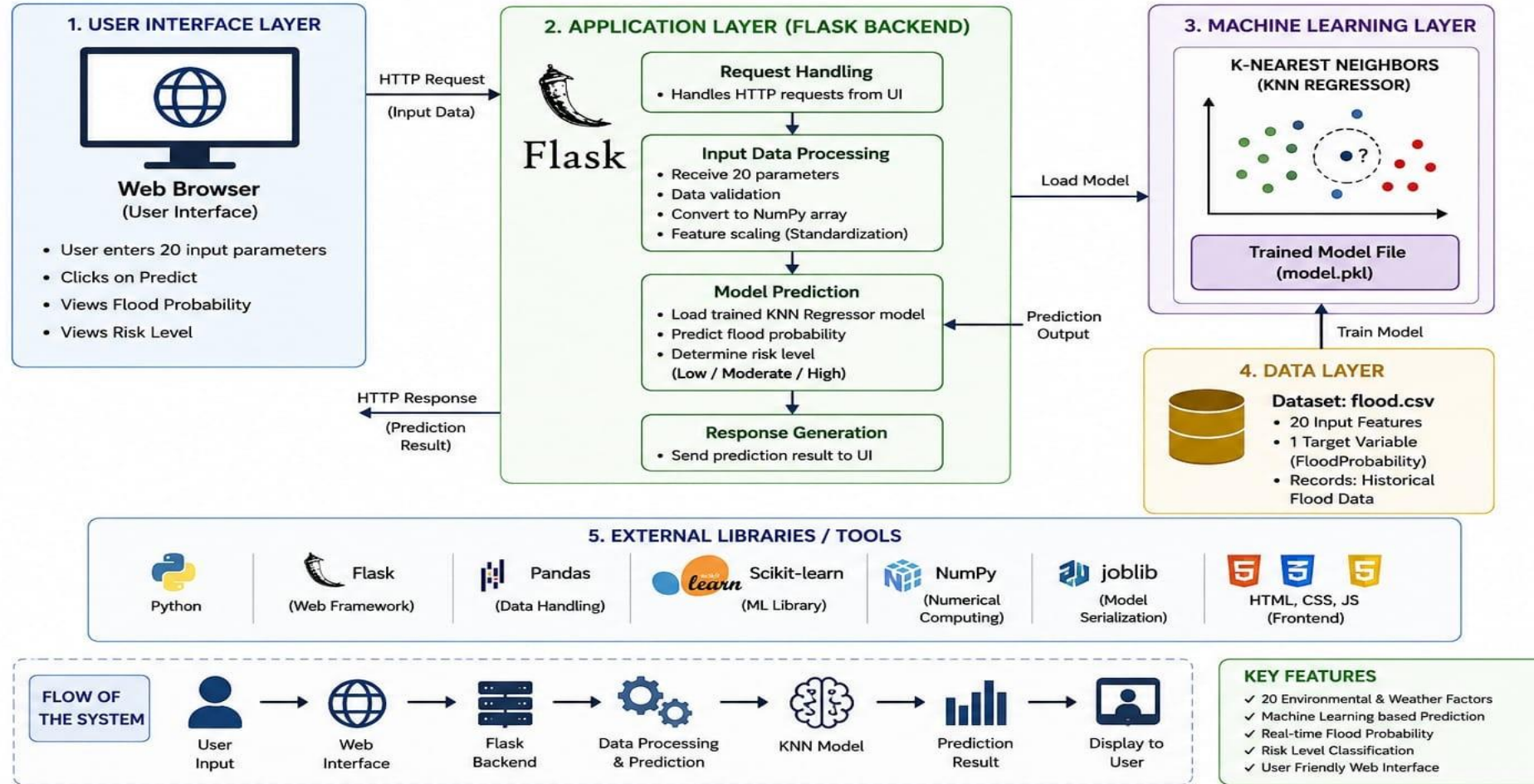
Outcome

The system provides a fast and accurate flood risk prediction, helping users understand the likelihood of flooding and supporting better disaster preparedness and decision-making.

TECHNICAL ARCHITECTURE

TECHNICAL ARCHITECTURE – RISING WATERS (FLOOD PREDICTION SYSTEM)

The system predicts flood probability using Machine Learning (KNN Regressor) based on 20 environmental and weather related input parameters.



SCREENSHOTS (PROJECT OUTPUT)

Home Page

 **Rising Waters**
Flood Prediction System

Enter values between **0** and **10** for all the parameters.

Monsoon Intensity (0-10)

Topography Drainage (0-10)

River Management (0-10)

Deforestation (0-10)

Urbanization (0-10)

Climate Change (0-10)

Dams Quality (0-10)

Siltation (0-10)

Agricultural Practices (0-10)

Encroachments (0-10)

Disaster Preparedness (0-10)

Drainage Systems (0-10)

Coastal Vulnerability (0-10)

Landslides (0-10)

Watersheds (0-10)

Deteriorating Infrastructure (0-10)

Population Score (0-10)

Wetland Loss (0-10)

Inadequate Planning (0-10)

Political Factors (0-10)

Predict Flood Probability



Prediction Result Page



Rising Waters

Flood Prediction Result

Flood Probability

0.337

Flood Risk Level

Moderate Flood Risk

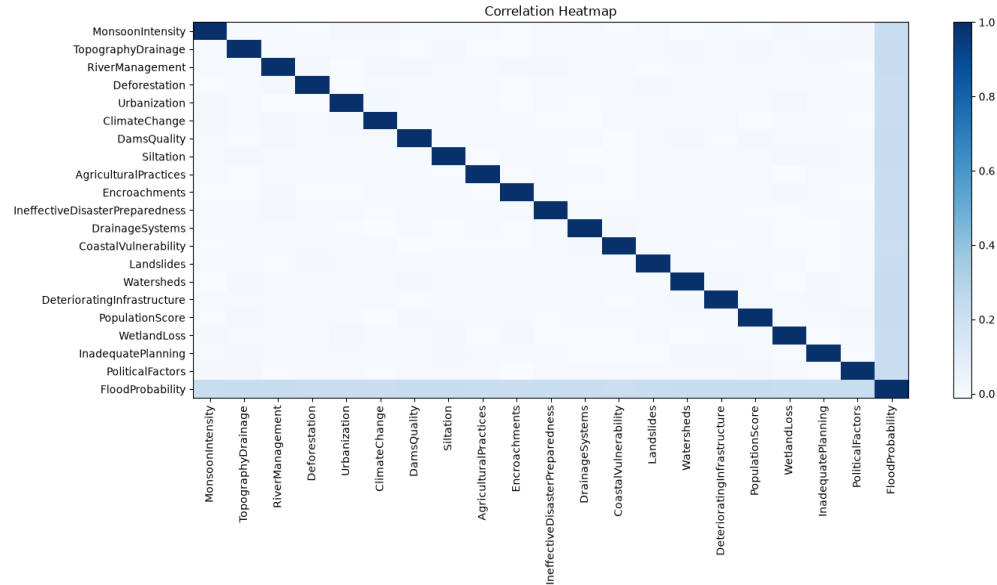
The predicted flood probability is moderate. Stay alert and monitor weather updates.



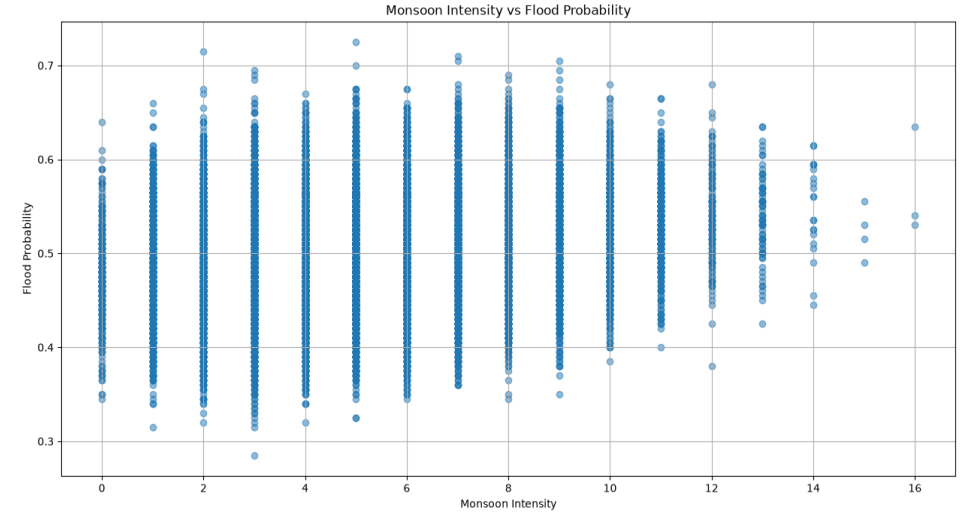
Predict Again



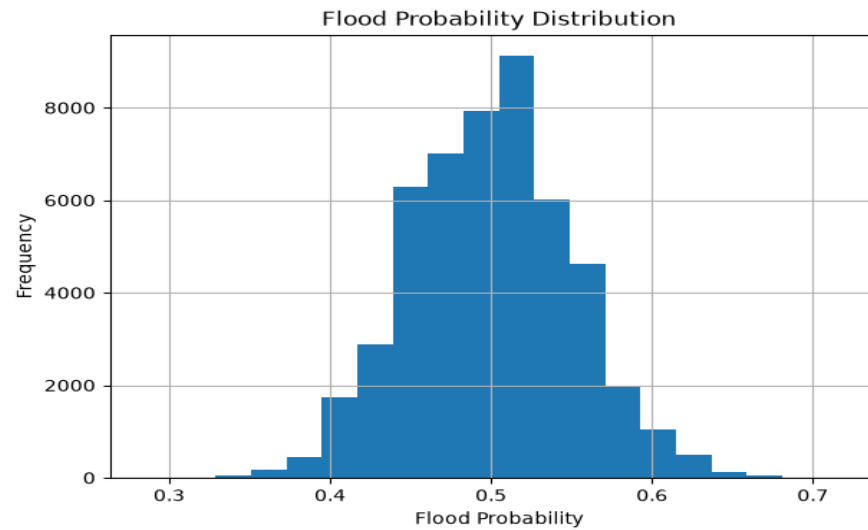
Correlation Heatmap



Scatter Plot



Flood Probability Distribution



SCREENSHOTS (PROJECT OUTPUT)

Home Page

- Displays the input form where users enter 20 environmental parameters.

Prediction Result Page

- Shows the predicted flood probability and the corresponding risk level (Low, Moderate, or High).

Correlation Heatmap

- Illustrates the relationship between different environmental factors and flood probability.

Histogram

- Displays the distribution of flood probability values in the dataset.

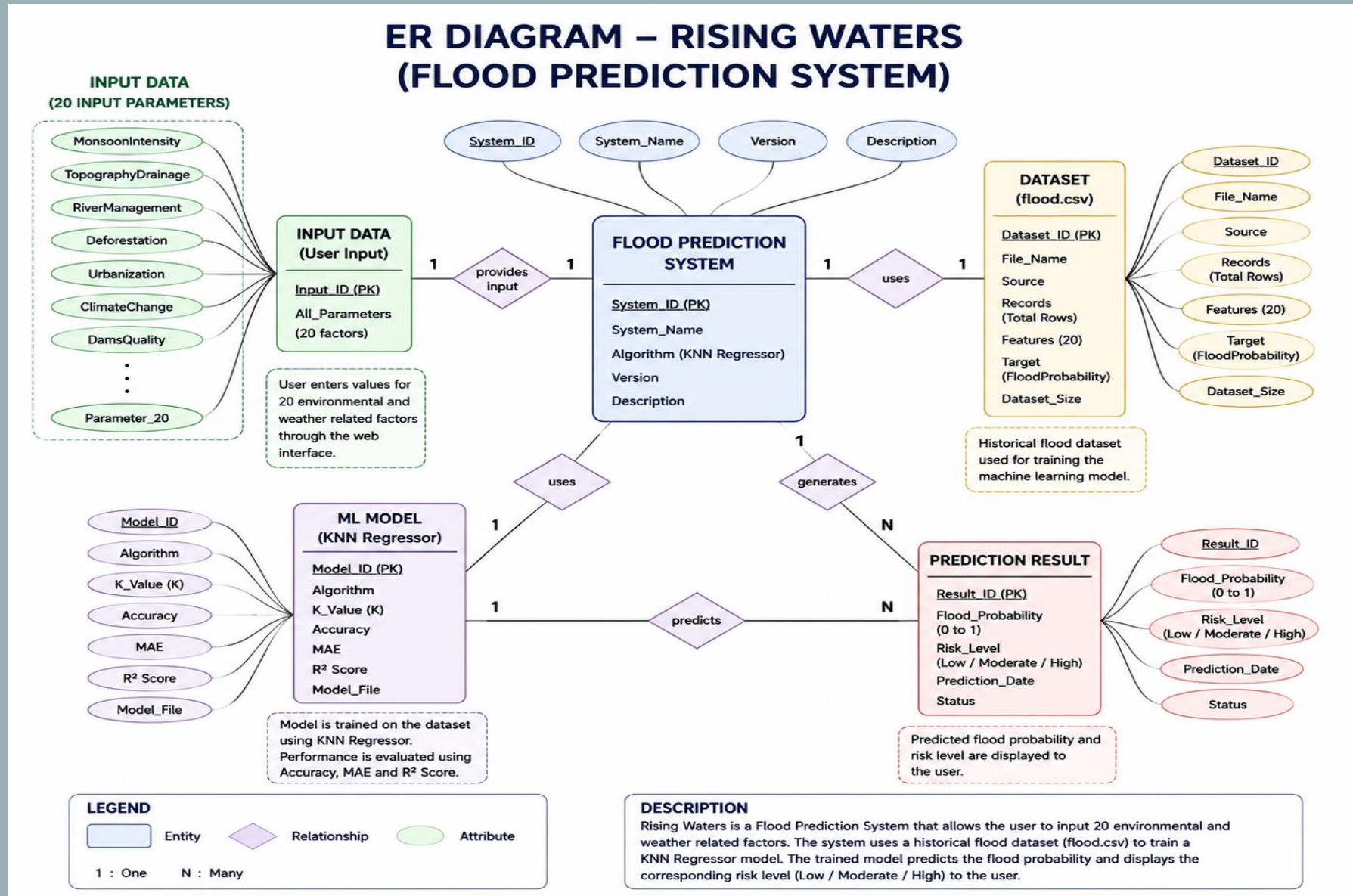
Scatter Plot

- Shows the relationship between Monsoon Intensity and Flood Probability.

Flood Probability Distribution

- Visualizes how flood probability values are distributed across the dataset.

ER DATASET



FUTURE SCOPE

- ❑ The **Rising Waters** project can be further enhanced by integrating advanced technologies and real-time data sources to improve prediction accuracy and support disaster management more effectively.
- ❑ Future improvements will make the system more scalable, intelligent, and beneficial for both government agencies and the public.

Future Enhancements

-  **Real-Time Weather Data Integration** using APIs to provide live flood predictions.
-  **IoT Sensor Integration** for monitoring river water levels and rainfall in real time.
-  **Mobile Application Development** for easy access to flood alerts and predictions.
-  **GIS-Based Flood Risk Mapping** to visualize flood-prone areas on interactive maps.
-  **Cloud Deployment** to allow users to access the application from anywhere.
-  **Advanced AI Models** such as Random Forest, XGBoost, or Deep Learning to improve prediction accuracy.
-  **SMS and Email Alert System** to notify users and emergency services during high flood-risk situations.
-  **Government and Disaster Management Integration** to support early warning systems and emergency response planning.

THANK YOU